Nexus vPC — Full Configuration v2

Nexus 1 + Nexus 2 | Exact Port Assignments from Your Document | Fortigate + Servers + PA

Netmagic Datacenter | Hassan | Draft — Verify before applying

CRITICAL DRAFT — All port numbers taken directly from your port assignment document. Verify physical cabling before applying.

Port Assignment Reference

```
! Topology layer summary:
! [SERVER1]          [SERVER2]
!     |                  |
!  vPC trunk          vPC trunk
!     |                  |
! [NEXUS 1] <--vPC--> [NEXUS 2]   (Keepalive: 10.10.10.1/2)
! LAN: 172.27.8.0/24
!     |                  |
!  FW LACP + FS MCLAG
!     |                  |
! [FGCORE1]         [FGCORE2]     (Fortigate Core L3)
!     |                  |
!  FW LACP + FS MCLAG
! LAN: 172.27.8.0/24
!     |                  |
! [FGSW1]           [FGSW2]       (Fortigate SW)
!     |                  |
!  FS MCLAG + PA LACP
!     |                  |
! [PA PARAM 1] [PA PARAM 2] (Palo Alto)
```

Port Assignment Plan

> ℹ Adjust these port numbers to match your actual Nexus hardware slot/port layout.

```
! ============================================================
! PORT ASSIGNMENT — NEXUS 1 & NEXUS 2 (symmetric)
! ============================================================
! Ethernet1/49-50      = vPC Peer-Link (ISL between Nexus1 <-> Nexus2 Po50)
! Ethernet1/47-48      = vPC Keep-alive (Po100, 10.10.10.1/2 Po 48)
! Ethernet1/46         = Fortigate Core1 (Po46 on Nexus1 / Po46 on Nexus2)
! Ethernet1/45         = Fortigate Core2 (Po45 on Nexus1 / Po45 on Nexus2)
! Ethernet1/4-5        = SERVER SW Po2 dual-home (Po104 on Nexus1 / Po104 on Nexus2)
! Ethernet1/6-7        = SERVER SW Po3 dual-home (Po105 on Nexus1 / Po105 on Nexus2)
! Ethernet1/8-9        = SERVER SW Po4 dual-home (Po106 on Nexus1 / Po106 on Nexus2)
! Ethernet1/10-11      = SERVER SW Po5 dual-home (Po107 on Nexus1 / Po107 on Nexus2)
! Ethernet1/12-13      = SERVER SW Po6 dual-home (Po108 on Nexus1 / Po108 on Nexus2)
! Ethernet1/14-15      = SERVER SW Po7 dual-home (Po109 on Nexus1 / Po109 on Nexus2)
! Ethernet1/16-17      = SERVER SW Po8 dual-home (Po110 on Nexus1 / Po110 on Nexus2)
! Ethernet1/18-19      = SERVER SW Po9 dual-home (Po111 on Nexus1 / Po111 on Nexus2)
! Ethernet1/20-21      = SERVER SW Po10 dual-home (Po112 on Nexus1 / Po112 on Nexus2)
! Ethernet1/22-23      = SERVER SW Po11 dual-home (Po113 on Nexus1 / Po113 on Nexus2)
! Loopback0            = NTP/SNMP source
! Loopback1            = 172.27.7.1/30 (replicated from CBS)
```

Section 1 — Feature Enables (Both Switches — Apply First)

> INFO Enable features BEFORE any interface or vPC config. 'feature vpc' must be present before configuring vpc domain.

NEXUS 1

```
hostname Netmagic-Nexus1

feature lacp
feature vpc
feature interface-vlan
feature lldp
feature ssh
feature snmp
feature hsrp
ip routing
no feature telnet
```

NEXUS 2

```
hostname Netmagic-Nexus2

feature lacp
feature vpc
feature interface-vlan
feature lldp
feature ssh
feature snmp
feature hsrp
ip routing
no feature telnet
```

Section 2 — vPC Domain, Dual Keepalive & Peer-Link (Po50)

CRITICAL Keepalive links (E1/47 + E1/48) MUST be UP and both IPs reachable BEFORE Po50 peer-link comes up. Bring up keepalives first, verify reachability, then bring up Po50.

INFO Dual keepalive: primary 10.10.10.x on E1/47, secondary 20.20.20.x on E1/48. NX-OS uses primary by default. Secondary is fallback.

WARN Mask corrected to /30 (255.255.255.252) on both sides — your document showed /24 on Nexus2 side which would overlap with other subnets.

NEXUS 1 — Keepalive Links (E1/47, E1/48)

```
interface Ethernet1/47
  description vPC-Keepalive-PRIMARY-10.10.10.x
  no switchport
  ip address 10.10.10.1 255.255.255.252
  no shutdown

interface Ethernet1/48
  description vPC-Keepalive-SECONDARY-20.20.20.x
  no switchport
  ip address 20.20.20.1 255.255.255.252
 no shutdown
```

NEXUS 1 — vPC Domain 10

```
vpc domain 10
  peer-keepalive destination 10.10.10.2 source 10.10.10.1
  peer-keepalive destination 20.20.20.2 source 20.20.20.1
  peer-gateway
  auto-recovery
  auto-recovery reload-delay 360
  delay restore 150
  ip arp synchronize
```

NEXUS 1 — Peer-Link Po50 (E1/49 + E1/50)

```
interface port-channel50
  description vPC-Peer-Link-ISL-to-Nexus2
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 1-39,45
  spanning-tree port type network
  vpc peer-link
  no shutdown

interface Ethernet1/49
  description vPC-ISL-to-Nexus2-MemberA
  switchport
  switchport mode trunk
  channel-group 50 mode active
  no shutdown
```

```
interface Ethernet1/50
  description vPC-ISL-to-Nexus2-MemberB
  switchport
  switchport mode trunk
  channel-group 50 mode active
  no shutdown
```

NEXUS 2 — Keepalive Links (E1/47, E1/48)

```
interface Ethernet1/47
  description vPC-Keepalive-PRIMARY-to-Nexus1
  no switchport
  ip address 10.10.10.2 255.255.255.252
  no shutdown

interface Ethernet1/48
  description vPC-Keepalive-SECONDARY-to-Nexus1
  no switchport
  ip address 20.20.20.2 255.255.255.252
  no shutdown
```

NEXUS 2 — vPC Domain 10

```
vpc domain 10
  peer-keepalive destination 10.10.10.1 source 10.10.10.2
  peer-keepalive destination 20.20.20.1 source 20.20.20.2
  peer-gateway
  auto-recovery
  auto-recovery reload-delay 360
  delay restore 150
  ip arp synchronize
```

NEXUS 2 — Peer-Link Po50 (E1/49 + E1/50)

```
interface port-channel50
  description vPC-Peer-Link-ISL-to-Nexus1
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 1-39,45
  spanning-tree port type network
  vpc peer-link
  no shutdown

interface Ethernet1/49
  description vPC-ISL-to-Nexus1-MemberA
  switchport
  switchport mode trunk
  channel-group 50 mode active
  no shutdown

interface Ethernet1/50
  description vPC-ISL-to-Nexus1-MemberB
```

```
switchport
switchport mode trunk
channel-group 50 mode active
no shutdown
```

Section 3 — VLAN Database (Identical on Both Switches)

```
! ==============================================================
! VLAN DATABASE
! ==============================================================
vlan 11
  name BASE-VLAN
vlan 12
  name DATABASE-VLAN
vlan 13
  name ANCILLARY-VLAN
vlan 14
  name HSM-VLAN
vlan 15
  name PMP-VLAN
vlan 16
  name NAS-VLAN
vlan 20
  name ATM-VLAN
vlan 21
  name CARD-MGMT-VLAN
vlan 22
  name ECOMMERCE-VLAN
vlan 23
  name WEB-SERVERS-VLAN
vlan 24
  name IMPS-SWITCH
vlan 25
  name IMPS-WEB
vlan 26
  name IMPS-EXTERNAL
vlan 27
  name UPI-SWITCH
vlan 28
  name UPI-WEB
vlan 29
  name UPI-EXTERNAL
vlan 30
  name BBPS-SWITCH
vlan 31
  name BBPS-WEB
vlan 32
  name BBPS-INTERNAL
vlan 33
  name FUTURE-USE
vlan 34
```

```
  name AEPS-EKYC
vlan 35
  name WEB-EKYC
vlan 36
  name CBS
vlan 37
  name BBPS-INTERNAL-2
vlan 38
  name FUTURE-USE-2
vlan 39
  name UAT-OOB
vlan 40
  name UAT
vlan 41
  name PAPG
vlan 45
 name L2-ONLY
```

Section 4 — Spanning Tree

> CRITICAL Set STP priority BEFORE any trunk comes up. Nexus 1 = primary root (4096), Nexus 2 = secondary root (8192).

NEXUS 1 — STP Primary Root

```
spanning-tree mode rapid-pvst
spanning-tree vlan 1-39,45 root primary
```

NEXUS 2 — STP Secondary Root

```
spanning-tree mode rapid-pvst
spanning-tree vlan 1-39,45 root secondary
```

Section 5 — Fortigate Core 1 & 2 (E1/46 → Po46, E1/45 → Po45)

> INFO Single physical port per Nexus per Fortigate (E1/46 for FGCore1, E1/45 for FGCore2). vPC provides redundancy across the two Nexus switches. If you need per-switch port-channel resilience, add a second port per Fortigate and bundle.

Fortigate Core 1 — Po46, vPC 46

```
! --- NEXUS 1 ---
interface port-channel46
  description FGCore1-vPC46
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 1-39,45
  vpc 46
  no shutdown

interface Ethernet1/46
  description FGCore1-Link-Nexus1
  switchport
  switchport mode trunk
  channel-group 46 mode active
  no shutdown

! --- NEXUS 2 ---
interface port-channel46
  description FGCore1-vPC46
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 1-39,45
  vpc 46
  no shutdown

interface Ethernet1/46
  description FGCore1-Link-Nexus2
  switchport
  switchport mode trunk
  channel-group 46 mode active
  no shutdown
```

Fortigate Core 2 — Po45, vPC 45

```
! --- NEXUS 1 ---
interface port-channel45
  description FGCore2-vPC45
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 1-39,45
  vpc 45
  no shutdown

interface Ethernet1/45
  description FGCore2-Link-Nexus1
```

```
  switchport
  switchport mode trunk
  channel-group 45 mode active
  no shutdown

! --- NEXUS 2 ---
interface port-channel45
  description FGCore2-vPC45
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 1-39,45
  vpc 45
  no shutdown

interface Ethernet1/45
  description FGCore2-Link-Nexus2
  switchport
  switchport mode trunk
  channel-group 45 mode active
  no shutdown
```

Section 6 — Server Port-Channels Po104–Po113 (E1/4–E1/23)

INFO Each server is dual-homed across Nexus 1 and Nexus 2. The same port-channel number and vPC number are used on both switches. Po110 and Po111 use lacp rate fast (was 'lacp timeout short' on CBS for FSNETNTX1127/1128).

WARN Server-side NIC bonding must be LACP (mode 4 / 802.3ad on Linux, LACP on VMware ESXi) before cutover. Verify before connecting.

FSNETNTX1121 — Po104, vPC 104 (Ethernet1/4 + Ethernet1/5)

```
! --- NEXUS 1 ---
interface port-channel104
  description FSNETNTX1121-vPC104
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 1-39,45
  vpc 104
  no shutdown

interface Ethernet1/4
  description FSNETNTX1121-Link1-Nexus1
  switchport
  switchport mode trunk
  channel-group 104 mode active
  no shutdown

interface Ethernet1/5
  description FSNETNTX1121-Link2-Nexus1
  switchport
  switchport mode trunk
```

```
  channel-group 104 mode active
  no shutdown

! --- NEXUS 2 ---
interface port-channel104
  description FSNETNTX1121-vPC104
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 1-39,45
  vpc 104
  no shutdown

interface Ethernet1/4
  description FSNETNTX1121-Link1-Nexus2
  switchport
  switchport mode trunk
  channel-group 104 mode active
  no shutdown

interface Ethernet1/5
  description FSNETNTX1121-Link2-Nexus2
  switchport
  switchport mode trunk
  channel-group 104 mode active
  no shutdown
```

FSNETNTX1122 — Po105, vPC 105 (Ethernet1/6 + Ethernet1/7)

```
! --- NEXUS 1 ---
interface port-channel105
  description FSNETNTX1122-vPC105
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 1-39,45
  vpc 105
  no shutdown

interface Ethernet1/6
  description FSNETNTX1122-Link1-Nexus1
  switchport
  switchport mode trunk
  channel-group 105 mode active
  no shutdown

interface Ethernet1/7
  description FSNETNTX1122-Link2-Nexus1
  switchport
  switchport mode trunk
  channel-group 105 mode active
  no shutdown

! --- NEXUS 2 ---
```

```
interface port-channel105
  description FSNETNTX1122-vPC105
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 1-39,45
  vpc 105
  no shutdown

interface Ethernet1/6
  description FSNETNTX1122-Link1-Nexus2
  switchport
  switchport mode trunk
  channel-group 105 mode active
  no shutdown

interface Ethernet1/7
  description FSNETNTX1122-Link2-Nexus2
  switchport
  switchport mode trunk
  channel-group 105 mode active
  no shutdown
```

FSNETNTX1123 — Po106, vPC 106 (Ethernet1/8 + Ethernet1/9)

```
! --- NEXUS 1 ---
interface port-channel106
  description FSNETNTX1123-vPC106
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 1-39,45
  vpc 106
  no shutdown

interface Ethernet1/8
  description FSNETNTX1123-Link1-Nexus1
  switchport
  switchport mode trunk
  channel-group 106 mode active
  no shutdown

interface Ethernet1/9
  description FSNETNTX1123-Link2-Nexus1
  switchport
  switchport mode trunk
  channel-group 106 mode active
  no shutdown

! --- NEXUS 2 ---
interface port-channel106
  description FSNETNTX1123-vPC106
  switchport
  switchport mode trunk
```

```
  switchport trunk allowed vlan 1-39,45
  vpc 106
  no shutdown

interface Ethernet1/8
  description FSNETNTX1123-Link1-Nexus2
  switchport
  switchport mode trunk
  channel-group 106 mode active
  no shutdown

interface Ethernet1/9
  description FSNETNTX1123-Link2-Nexus2
  switchport
  switchport mode trunk
  channel-group 106 mode active
  no shutdown
```

FSNETNTX1124 — Po107, vPC 107 (Ethernet1/10 + Ethernet1/11)

```
! --- NEXUS 1 ---
interface port-channel107
  description FSNETNTX1124-vPC107
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 1-39,45
  vpc 107
  no shutdown

interface Ethernet1/10
  description FSNETNTX1124-Link1-Nexus1
  switchport
  switchport mode trunk
  channel-group 107 mode active
  no shutdown

interface Ethernet1/11
  description FSNETNTX1124-Link2-Nexus1
  switchport
  switchport mode trunk
  channel-group 107 mode active
  no shutdown

! --- NEXUS 2 ---
interface port-channel107
  description FSNETNTX1124-vPC107
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 1-39,45
  vpc 107
  no shutdown
```

```
interface Ethernet1/10
  description FSNETNTX1124-Link1-Nexus2
  switchport
  switchport mode trunk
  channel-group 107 mode active
  no shutdown

interface Ethernet1/11
  description FSNETNTX1124-Link2-Nexus2
  switchport
  switchport mode trunk
  channel-group 107 mode active
  no shutdown
```

FSNETNTX1125 — Po108, vPC 108 (Ethernet1/12 + Ethernet1/13)

```
! --- NEXUS 1 ---
interface port-channel108
  description FSNETNTX1125-vPC108
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 1-39,45
  vpc 108
  no shutdown

interface Ethernet1/12
  description FSNETNTX1125-Link1-Nexus1
  switchport
  switchport mode trunk
  channel-group 108 mode active
  no shutdown

interface Ethernet1/13
  description FSNETNTX1125-Link2-Nexus1
  switchport
  switchport mode trunk
  channel-group 108 mode active
  no shutdown

! --- NEXUS 2 ---
interface port-channel108
  description FSNETNTX1125-vPC108
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 1-39,45
  vpc 108
  no shutdown

interface Ethernet1/12
  description FSNETNTX1125-Link1-Nexus2
  switchport
  switchport mode trunk
```

```
  channel-group 108 mode active
  no shutdown

interface Ethernet1/13
  description FSNETNTX1125-Link2-Nexus2
  switchport
  switchport mode trunk
  channel-group 108 mode active
  no shutdown
```

FSNETNTX1126 — Po109, vPC 109 (Ethernet1/14 + Ethernet1/15)

```
! --- NEXUS 1 ---
interface port-channel109
  description FSNETNTX1126-vPC109
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 1-39,45
  vpc 109
  no shutdown

interface Ethernet1/14
  description FSNETNTX1126-Link1-Nexus1
  switchport
  switchport mode trunk
  channel-group 109 mode active
  no shutdown

interface Ethernet1/15
  description FSNETNTX1126-Link2-Nexus1
  switchport
  switchport mode trunk
  channel-group 109 mode active
  no shutdown

! --- NEXUS 2 ---
interface port-channel109
  description FSNETNTX1126-vPC109
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 1-39,45
  vpc 109
  no shutdown

interface Ethernet1/14
  description FSNETNTX1126-Link1-Nexus2
  switchport
  switchport mode trunk
  channel-group 109 mode active
  no shutdown

interface Ethernet1/15
```

```
  description FSNETNTX1126-Link2-Nexus2
  switchport
  switchport mode trunk
  channel-group 109 mode active
  no shutdown
```

FSNETNTX1127 — Po110, vPC 110 (Ethernet1/16 + Ethernet1/17)

```
! --- NEXUS 1 ---
interface port-channel110
  description FSNETNTX1127-vPC110
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 1-39,45
  vpc 110
  no shutdown

interface Ethernet1/16
  description FSNETNTX1127-Link1-Nexus1
  switchport
  switchport mode trunk
  lacp rate fast
  channel-group 110 mode active
  no shutdown

interface Ethernet1/17
  description FSNETNTX1127-Link2-Nexus1
  switchport
  switchport mode trunk
  lacp rate fast
  channel-group 110 mode active
  no shutdown

! --- NEXUS 2 ---
interface port-channel110
  description FSNETNTX1127-vPC110
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 1-39,45
  vpc 110
  no shutdown

interface Ethernet1/16
  description FSNETNTX1127-Link1-Nexus2
  switchport
  switchport mode trunk
  lacp rate fast
  channel-group 110 mode active
  no shutdown

interface Ethernet1/17
  description FSNETNTX1127-Link2-Nexus2
```

```
  switchport
  switchport mode trunk
  lacp rate fast
  channel-group 110 mode active
  no shutdown
```

FSNETNTX1128 — Po111, vPC 111 (Ethernet1/18 + Ethernet1/19)

```
! --- NEXUS 1 ---
interface port-channel111
  description FSNETNTX1128-vPC111
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 1-39,45
  vpc 111
  no shutdown

interface Ethernet1/18
  description FSNETNTX1128-Link1-Nexus1
  switchport
  switchport mode trunk
  lacp rate fast
  channel-group 111 mode active
  no shutdown

interface Ethernet1/19
  description FSNETNTX1128-Link2-Nexus1
  switchport
  switchport mode trunk
  lacp rate fast
  channel-group 111 mode active
  no shutdown

! --- NEXUS 2 ---
interface port-channel111
  description FSNETNTX1128-vPC111
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 1-39,45
  vpc 111
  no shutdown

interface Ethernet1/18
  description FSNETNTX1128-Link1-Nexus2
  switchport
  switchport mode trunk
  lacp rate fast
  channel-group 111 mode active
  no shutdown

interface Ethernet1/19
  description FSNETNTX1128-Link2-Nexus2
```

```
  switchport
  switchport mode trunk
  lacp rate fast
  channel-group 111 mode active
  no shutdown
```

SERVER-SW-9 — Po112, vPC 112 (Ethernet1/20 + Ethernet1/21)

```
! --- NEXUS 1 ---
interface port-channel112
  description SERVER-SW-9-vPC112
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 1-39,45
  vpc 112
  no shutdown

interface Ethernet1/20
  description SERVER-SW-9-Link1-Nexus1
  switchport
  switchport mode trunk
  channel-group 112 mode active
  no shutdown

interface Ethernet1/21
  description SERVER-SW-9-Link2-Nexus1
  switchport
  switchport mode trunk
  channel-group 112 mode active
  no shutdown

! --- NEXUS 2 ---
interface port-channel112
  description SERVER-SW-9-vPC112
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 1-39,45
  vpc 112
  no shutdown

interface Ethernet1/20
  description SERVER-SW-9-Link1-Nexus2
  switchport
  switchport mode trunk
  channel-group 112 mode active
  no shutdown

interface Ethernet1/21
  description SERVER-SW-9-Link2-Nexus2
  switchport
  switchport mode trunk
  channel-group 112 mode active
```

```
  no shutdown
```

SERVER-SW-10 — Po113, vPC 113 (Ethernet1/22 + Ethernet1/23)

```
! --- NEXUS 1 ---
interface port-channel113
  description SERVER-SW-10-vPC113
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 1-39,45
  vpc 113
  no shutdown

interface Ethernet1/22
  description SERVER-SW-10-Link1-Nexus1
  switchport
  switchport mode trunk
  channel-group 113 mode active
  no shutdown

interface Ethernet1/23
  description SERVER-SW-10-Link2-Nexus1
  switchport
  switchport mode trunk
  channel-group 113 mode active
  no shutdown

! --- NEXUS 2 ---
interface port-channel113
  description SERVER-SW-10-vPC113
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 1-39,45
  vpc 113
  no shutdown

interface Ethernet1/22
  description SERVER-SW-10-Link1-Nexus2
  switchport
  switchport mode trunk
  channel-group 113 mode active
  no shutdown

interface Ethernet1/23
  description SERVER-SW-10-Link2-Nexus2
  switchport
  switchport mode trunk
  channel-group 113 mode active
  no shutdown
```

Section 7 — SVI Routing + HSRP (All 31 VLANs)

INFO Nexus 1 = HSRP active (priority 110 + preempt) for all VLANs. Nexus 2 = standby (priority 100). HSRP virtual IP (.1) is what all downstream devices use as their default gateway. With peer-gateway enabled, both Nexus switches can forward regardless of HSRP state.

WARN HSRP virtual IPs (.1 per subnet) replace the old CBS SVI IPs (.1 per subnet) — the gateway address hosts use is unchanged. Only the physical Nexus IPs (.2 and .3) are new.

CRITICAL Update Fortigate and Palo Alto interface gateway config to use .1 virtual IPs (which is what the CBS had). No change needed on hosts — same gateway.

NEXUS 1 — All SVIs (HSRP Active)

```
! ============================================================
! SVIs + HSRP - NEXUS 1 (Active, priority 110, preempt)
! Physical IP: .2 | HSRP Virtual: .1
! ============================================================

interface Vlan1
  ip address 172.27.8.30 255.255.255.0
  hsrp version 2
  hsrp 1
    ip 172.27.8.1
    priority 110
    preempt
  no shutdown

interface Vlan10
  ip address 172.27.10.2 255.255.255.0
  hsrp version 2
  hsrp 10
    ip 172.27.10.1
    priority 110
    preempt
  no shutdown

interface Vlan11
  ip address 172.27.11.2 255.255.255.0
  hsrp version 2
  hsrp 11
    ip 172.27.11.1
    priority 110
    preempt
  no shutdown

interface Vlan12
  ip address 172.27.12.2 255.255.255.0
  hsrp version 2
  hsrp 12
    ip 172.27.12.1
    priority 110
    preempt
  no shutdown
```

```
interface Vlan13
  ip address 172.27.13.2 255.255.255.0
  ipv6 address 2401:8800:c13:2::85/126
  hsrp version 2
  hsrp 13
    ip 172.27.13.1
    priority 110
    preempt
  no shutdown

interface Vlan14
  ip address 172.27.14.2 255.255.255.0
  hsrp version 2
  hsrp 14
    ip 172.27.14.1
    priority 110
    preempt
  no shutdown

interface Vlan15
  ip address 172.27.15.2 255.255.255.0
  hsrp version 2
  hsrp 15
    ip 172.27.15.1
    priority 110
    preempt
  no shutdown

interface Vlan16
  ip address 172.27.16.2 255.255.255.0
  hsrp version 2
  hsrp 16
    ip 172.27.16.1
    priority 110
    preempt
  no shutdown

interface Vlan17
  ip address 172.27.17.2 255.255.255.0
  hsrp version 2
  hsrp 17
    ip 172.27.17.1
    priority 110
    preempt
  no shutdown

interface Vlan18
  ip address 172.27.18.2 255.255.255.0
  hsrp version 2
  hsrp 18
    ip 172.27.18.1
    priority 110
```

```
    preempt
  no shutdown

interface Vlan19
  ip address 172.27.19.2 255.255.255.0
  hsrp version 2
  hsrp 19
    ip 172.27.19.1
    priority 110
    preempt
  no shutdown

interface Vlan20
  ip address 172.27.20.2 255.255.255.0
  hsrp version 2
  hsrp 20
    ip 172.27.20.1
    priority 110
    preempt
  no shutdown

interface Vlan21
  ip address 172.27.21.2 255.255.255.0
  hsrp version 2
  hsrp 21
    ip 172.27.21.1
    priority 110
    preempt
  no shutdown

interface Vlan22
  ip address 172.27.22.2 255.255.255.0
  hsrp version 2
  hsrp 22
    ip 172.27.22.1
    priority 110
    preempt
  no shutdown

interface Vlan23
  ip address 172.27.23.2 255.255.255.0
  ipv6 enable
  hsrp version 2
  hsrp 23
    ip 172.27.23.1
    priority 110
    preempt
  no shutdown

interface Vlan24
  ip address 172.27.24.2 255.255.255.0
  hsrp version 2
```

```
  hsrp 24
    ip 172.27.24.1
    priority 110
    preempt
  no shutdown

interface Vlan25
  ip address 172.27.25.2 255.255.255.0
  hsrp version 2
  hsrp 25
    ip 172.27.25.1
    priority 110
    preempt
  no shutdown

interface Vlan26
  ip address 172.27.26.2 255.255.255.0
  hsrp version 2
  hsrp 26
    ip 172.27.26.1
    priority 110
    preempt
  no shutdown

interface Vlan27
  ip address 172.27.27.2 255.255.255.0
  ipv6 enable
  hsrp version 2
  hsrp 27
    ip 172.27.27.1
    priority 110
    preempt
  no shutdown

interface Vlan28
  ip address 172.27.28.2 255.255.255.0
  hsrp version 2
  hsrp 28
    ip 172.27.28.1
    priority 110
    preempt
  no shutdown

interface Vlan29
  ip address 172.27.29.2 255.255.255.0
  hsrp version 2
  hsrp 29
    ip 172.27.29.1
    priority 110
    preempt
  no shutdown
```

```
interface Vlan30
  ip address 172.27.30.2 255.255.255.0
  hsrp version 2
  hsrp 30
    ip 172.27.30.1
    priority 110
    preempt
  no shutdown

interface Vlan31
  ip address 172.27.31.2 255.255.255.0
  hsrp version 2
  hsrp 31
    ip 172.27.31.1
    priority 110
    preempt
  no shutdown

interface Vlan32
  ip address 172.27.32.2 255.255.255.0
  hsrp version 2
  hsrp 32
    ip 172.27.32.1
    priority 110
    preempt
  no shutdown

interface Vlan33
  ip address 172.27.33.2 255.255.255.0
  hsrp version 2
  hsrp 33
    ip 172.27.33.1
    priority 110
    preempt
  no shutdown

interface Vlan34
  ip address 172.27.34.2 255.255.255.0
  hsrp version 2
  hsrp 34
    ip 172.27.34.1
    priority 110
    preempt
  no shutdown

interface Vlan35
  ip address 172.27.35.2 255.255.255.0
  hsrp version 2
  hsrp 35
    ip 172.27.35.1
    priority 110
    preempt
```

```
  no shutdown

interface Vlan36
  ip address 172.27.36.2 255.255.255.0
  hsrp version 2
  hsrp 36
    ip 172.27.36.1
    priority 110
    preempt
  no shutdown

interface Vlan37
  ip address 172.27.37.2 255.255.255.0
  ipv6 address 2401:8800:c13:2::91/124
  hsrp version 2
  hsrp 37
    ip 172.27.37.1
    priority 110
    preempt
  no shutdown

interface Vlan38
  ip address 172.27.38.2 255.255.255.0
  hsrp version 2
  hsrp 38
    ip 172.27.38.1
    priority 110
    preempt
  no shutdown

interface Vlan39
  ip address 172.27.39.2 255.255.255.0
  hsrp version 2
  hsrp 39
    ip 172.27.39.1
    priority 110
    preempt
  no shutdown

! VLAN 45 - L2 only, no SVI

ip default-gateway 172.27.8.1
```

NEXUS 2 — All SVIs (HSRP Standby)

```
! ============================================================
! SVIs + HSRP - NEXUS 2 (Standby, priority 100)
! Physical IP: .3 | HSRP Virtual: .1 (same as Nexus1)
! ============================================================

interface Vlan1
  ip address 172.27.8.31 255.255.255.0
```

```
  hsrp version 2
  hsrp 1
    ip 172.27.8.1
    priority 100
  no shutdown

interface Vlan10
  ip address 172.27.10.3 255.255.255.0
  hsrp version 2
  hsrp 10
    ip 172.27.10.1
    priority 100
  no shutdown

interface Vlan11
  ip address 172.27.11.3 255.255.255.0
  hsrp version 2
  hsrp 11
    ip 172.27.11.1
    priority 100
  no shutdown

interface Vlan12
  ip address 172.27.12.3 255.255.255.0
  hsrp version 2
  hsrp 12
    ip 172.27.12.1
    priority 100
  no shutdown

interface Vlan13
  ip address 172.27.13.3 255.255.255.0
  ipv6 address 2401:8800:c13:2::85/126
  hsrp version 2
  hsrp 13
    ip 172.27.13.1
    priority 100
  no shutdown

interface Vlan14
  ip address 172.27.14.3 255.255.255.0
  hsrp version 2
  hsrp 14
    ip 172.27.14.1
    priority 100
  no shutdown

interface Vlan15
  ip address 172.27.15.3 255.255.255.0
  hsrp version 2
  hsrp 15
    ip 172.27.15.1
```

```
    priority 100
  no shutdown

interface Vlan16
  ip address 172.27.16.3 255.255.255.0
  hsrp version 2
  hsrp 16
    ip 172.27.16.1
    priority 100
  no shutdown

interface Vlan17
  ip address 172.27.17.3 255.255.255.0
  hsrp version 2
  hsrp 17
    ip 172.27.17.1
    priority 100
  no shutdown

interface Vlan18
  ip address 172.27.18.3 255.255.255.0
  hsrp version 2
  hsrp 18
    ip 172.27.18.1
    priority 100
  no shutdown

interface Vlan19
  ip address 172.27.19.3 255.255.255.0
  hsrp version 2
  hsrp 19
    ip 172.27.19.1
    priority 100
  no shutdown

interface Vlan20
  ip address 172.27.20.3 255.255.255.0
  hsrp version 2
  hsrp 20
    ip 172.27.20.1
    priority 100
  no shutdown

interface Vlan21
  ip address 172.27.21.3 255.255.255.0
  hsrp version 2
  hsrp 21
    ip 172.27.21.1
    priority 100
  no shutdown

interface Vlan22
```

```
  ip address 172.27.22.3 255.255.255.0
  hsrp version 2
  hsrp 22
    ip 172.27.22.1
    priority 100
  no shutdown

interface Vlan23
  ip address 172.27.23.3 255.255.255.0
  ipv6 enable
  hsrp version 2
  hsrp 23
    ip 172.27.23.1
    priority 100
  no shutdown

interface Vlan24
  ip address 172.27.24.3 255.255.255.0
  hsrp version 2
  hsrp 24
    ip 172.27.24.1
    priority 100
  no shutdown

interface Vlan25
  ip address 172.27.25.3 255.255.255.0
  hsrp version 2
  hsrp 25
    ip 172.27.25.1
    priority 100
  no shutdown

interface Vlan26
  ip address 172.27.26.3 255.255.255.0
  hsrp version 2
  hsrp 26
    ip 172.27.26.1
    priority 100
  no shutdown

interface Vlan27
  ip address 172.27.27.3 255.255.255.0
  ipv6 enable
  hsrp version 2
  hsrp 27
    ip 172.27.27.1
    priority 100
  no shutdown

interface Vlan28
  ip address 172.27.28.3 255.255.255.0
  hsrp version 2
```

```
  hsrp 28
    ip 172.27.28.1
    priority 100
  no shutdown

interface Vlan29
  ip address 172.27.29.3 255.255.255.0
  hsrp version 2
  hsrp 29
    ip 172.27.29.1
    priority 100
  no shutdown

interface Vlan30
  ip address 172.27.30.3 255.255.255.0
  hsrp version 2
  hsrp 30
    ip 172.27.30.1
    priority 100
  no shutdown

interface Vlan31
  ip address 172.27.31.3 255.255.255.0
  hsrp version 2
  hsrp 31
    ip 172.27.31.1
    priority 100
  no shutdown

interface Vlan32
  ip address 172.27.32.3 255.255.255.0
  hsrp version 2
  hsrp 32
    ip 172.27.32.1
    priority 100
  no shutdown

interface Vlan33
  ip address 172.27.33.3 255.255.255.0
  hsrp version 2
  hsrp 33
    ip 172.27.33.1
    priority 100
  no shutdown

interface Vlan34
  ip address 172.27.34.3 255.255.255.0
  hsrp version 2
  hsrp 34
    ip 172.27.34.1
    priority 100
  no shutdown
```

```
interface Vlan35
  ip address 172.27.35.3 255.255.255.0
  hsrp version 2
  hsrp 35
    ip 172.27.35.1
    priority 100
  no shutdown

interface Vlan36
  ip address 172.27.36.3 255.255.255.0
  hsrp version 2
  hsrp 36
    ip 172.27.36.1
    priority 100
  no shutdown

interface Vlan37
  ip address 172.27.37.3 255.255.255.0
  ipv6 address 2401:8800:c13:2::91/124
  hsrp version 2
  hsrp 37
    ip 172.27.37.1
    priority 100
  no shutdown

interface Vlan38
  ip address 172.27.38.3 255.255.255.0
  hsrp version 2
  hsrp 38
    ip 172.27.38.1
    priority 100
  no shutdown

interface Vlan39
  ip address 172.27.39.3 255.255.255.0
  hsrp version 2
  hsrp 39
    ip 172.27.39.1
    priority 100
  no shutdown

! VLAN 45 - L2 only, no SVI

ip default-gateway 172.27.8.1
```

Section 8 — Loopback Interfaces

NEXUS 1

```
interface loopback0
  description NTP-SNMP-Source-Nexus1
  ip address 172.27.7.1 255.255.255.252
  no shutdown

interface loopback1
  description Replicated-CBS
  ip address 172.27.7.5 255.255.255.252
  no shutdown
```

NEXUS 2

```
interface loopback0
  description NTP-SNMP-Source-Nexus2
  ip address 172.27.7.2 255.255.255.252
  no shutdown

interface loopback1
  description Replicated-CBS
  ip address 172.27.7.6 255.255.255.252
  no shutdown
```

Section 9 — Management Plane (Both Switches)

NTP

```
ntp server 172.27.13.11 use-vrf default
ntp server 172.27.13.12 use-vrf default
ntp source-interface loopback0
clock timezone IST 5 30
```

Syslog

```
logging server 172.27.13.13
logging server 172.27.13.14
logging server 172.27.13.29
logging level default 6
logging console 2
logging timestamp milliseconds
logging source-interface loopback0
```

SNMP

```
snmp-server community FSNET ro
snmp-server host 172.27.13.14 version 2c FSNET
snmp-server host 172.27.13.19 version 2c FSNET
snmp-server host 172.27.13.191 version 2c FSNET
snmp-server location "Netmagic Datacenter"
```

```
snmp-server contact "Nikhil Kadam"
```

Users, SSH & Banner

CRITICAL Replace <ENCRYPTED> with actual encrypted hashes. Never store plaintext passwords in config.

```
username cnfmgr    password 5 <ENCRYPTED> role network-admin
username fsnetrw   password 5 <ENCRYPTED> role network-admin
username nikhil.k  password 5 <ENCRYPTED> role network-admin
username ajinkya.b password 5 <ENCRYPTED> role network-admin

feature ssh
ssh key rsa 2048
no feature telnet

line console
  exec-timeout 5

line vty
  exec-timeout 5
  transport input ssh

banner motd #
AUTHORIZED ACCESS ONLY
DISCONNECT ASAP IF YOU ARE NOT AUTHORISED
ACCESS MONITORED AND RECORDED
#
```

Section 10 — Post-Cutover Verification

INFO Run in order. vPC must be healthy before connecting any downstream device.

```
! Step 1 - vPC health (run FIRST on both switches)
show vpc
show vpc brief
show vpc peer-keepalive
show vpc consistency-parameters global
show vpc role

! Step 2 - Port-channels
show port-channel summary
show lacp neighbor
show lacp counters

! Step 3 - STP root confirmation
show spanning-tree summary
show spanning-tree vlan 1 detail

! Step 4 - SVIs and HSRP
show ip interface brief
show hsrp brief
```

```
show ip route

! Step 5 - VLAN membership
show vlan brief

! Step 6 - NTP
show ntp peer-status
show clock

! Step 7 - Connectivity
ping 172.27.8.31 source-interface Vlan1
ping 172.27.13.13 source-interface Vlan13
ping 172.27.13.11 source-interface loopback0

! Step 8 - LLDP neighbors
show lldp neighbors
show lldp neighbors detail

! Step 9 - Transceiver health
show interface transceiver

! Step 10 - Error counters baseline
show interface counters errors
```